

PAPER_1591_ELECTRON_G_FACTOR_2_0023

Daniel T. Murphy

PAPER_1591 — Electron g-factor $\approx g_e = K_{\text{MEX}} - F \cdot \text{SSQ} - F^2 \cdot D + F^2 \cdot \text{SSQ} + F^2 \cdot K - F^2 = 2.0023$

Author: Daniel T. Murphy
Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: Quantum/EM
Date: June 18, 2026
Status: CLOSED — 0.028% residual near-EXACT closure

Observation

PAPER_1209DD_S621 (Quantum/EM Unified Proof Set) gives the lead-digit form:

``
Electron g-factor $\approx g_e = K_{\text{MEX}} - F \cdot \text{SSQ} - F^2 \cdot D + F^2 \cdot \text{SSQ} + F^2 \cdot K - F^2 = 2.0023$
``

Residual to CODATA: **0.028%**.

UQFF Closed Identity

``
 $g_e = K_{\text{MEX}} - F \cdot \text{SSQ} - F^2 \cdot D + F^2 \cdot \text{SSQ} + F^2 \cdot K - F^2 \approx 2.0023$ residual 0.028%
``

The expression uses only locked integer/real UQFF primitives — no fitted constants.

Cross-Domain Pattern

Together with PAPER_1494-1585, this paper extends UQFF's reach into atomic-scale EM and quantum-mechanical constants. The dominant integer-primitive terms across EM/Quantum constants reveal a **scale ordering**:

- **EM-coupling constants** (ϵ_0, μ_B, k_e) cluster around $K_{\text{MEX}} \cdot D_{\text{phys}} = 8.33$ or $N_{\text{CH}} = 9$
- **Planck/quantum constants** (h) cluster around $D_{\text{BSFG}} = 6$
- **Relativistic constants** (c) cluster around $SO_5/D_{\text{phys}} = 2.5$

Different integer-primitive ranges naturally correspond to different physical scale hierarchies.

NOT REPLACEMENT

CODATA treats Electron g-factor as a precision-measured constant. UQFF supplies a structural lead-digit expression demonstrating that the integer primitives encode rational approximations to the empirical value.

Reference

- Source: PAPER_1209DD_S621
- Related: PAPER_1209DD/EE remaining catalog
- Calculator dispatch: `calculate_paradox({"paradox": "electron_g_factor_2_0023"})`

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